

External Tables Section LAB 1

```
USE DATABASE HRMS;
```

```
USE SCHEMA HR;
```

```
CREATE OR REPLACE FILE FORMAT HRMS.HR.CSV_ET_FILEFORMAT
TYPE = CSV
FIELD_DELIMITER = ','
SKIP_HEADER = 0
NULL_IF = ('NULL','Null','null')
FIELD_OPTIONALLY_ENCLOSED_BY = ''
EMPTY_FIELD_AS_NULL = TRUE;
```

```
CREATE OR REPLACE STAGE HRMS.HR.AWS_ET_CSV_STAGE
STORAGE_INTEGRATION = AWS_S3_INT
URL = 's3://learn2cloud-snowflake/external_table/csv/'
FILE_FORMAT = CSV_ET_FILEFORMAT;
```

```
LIST @AWS_ET_CSV_STAGE;
```

```
SELECT
    T.$1,
    T.$2,
    T.$3,
    T.$4,
    T.$5,
    T.$6 ,
    T.$7,
    T.$8,
    T.$9,
    T.$10,
    T.$11
FROM @AWS_ET_CSV_STAGE T;
```

```
CREATE OR REPLACE EXTERNAL TABLE EXT_EMPLOYEES
(
```

```
EMPLOYEE_ID  VARCHAR  AS (VALUE:c1::VARCHAR),
FIRST_NAME   VARCHAR  AS (VALUE:c2::VARCHAR),
LAST_NAME    VARCHAR  AS (VALUE:c3::VARCHAR),
EMAIL        VARCHAR  AS (VALUE:c4::VARCHAR),
PHONE_NUMBER VARCHAR  AS (VALUE:c5::VARCHAR),
HIRE_DATE    VARCHAR  AS (VALUE:c6::VARCHAR),
JOB_ID       VARCHAR  AS (VALUE:c7::VARCHAR),
SALARY       VARCHAR  AS (VALUE:c8::VARCHAR),
COMMISSION_PCT VARCHAR  AS (VALUE:c9::VARCHAR),
MANAGER_ID   VARCHAR  AS (VALUE:c10::VARCHAR),
DEPARTMENT_ID VARCHAR  AS (VALUE:c11::VARCHAR)
)
LOCATION=@AWS_ET_CSV_STAGE
PATTERN='*.employees.*[.]csv'
FILE_FORMAT = CSV_ET_FILEFORMAT;
```

```
DESCRIBE EXTERNAL TABLE EXT_EMPLOYEES;
```

```
SELECT * FROM EXT_EMPLOYEES;
```

```
SELECT METADATA$FILENAME,METADATA$FILE_ROW_NUMBER FROM
EXT_EMPLOYEES;
```

```
SELECT DISTINCT METADATA$FILENAME FROM EXT_EMPLOYEES;
```

```
SELECT METADATA$FILENAME,MAX(METADATA$FILE_ROW_NUMBER) FROM
EXT_EMPLOYEES
GROUP BY METADATA$FILENAME
ORDER BY 2 DESC;
```

```
SELECT * FROM
TABLE(INFORMATION_SCHEMA.EXTERNAL_TABLE_FILES('EXT_EMPLOYEES'));
```

```
SELECT * FROM
TABLE(INFORMATION_SCHEMA.EXTERNAL_TABLE_FILE_REGISTRATION_HISTORY('EXT
_EMPLOYEES'));
```

```
SELECT * FROM EXT_EMPLOYEES;
```

```
CREATE OR REPLACE EXTERNAL TABLE EXT_EMPLOYEES_FORMAT_DATE
(
  EMPLOYEE_ID  VARCHAR AS (VALUE:c1::VARCHAR),
  FIRST_NAME   VARCHAR AS (VALUE:c2::VARCHAR),
  LAST_NAME    VARCHAR AS (VALUE:c3::VARCHAR),
  EMAIL        VARCHAR AS (VALUE:c4::VARCHAR),
  PHONE_NUMBER VARCHAR AS (VALUE:c5::VARCHAR),
  HIRE_DATE    VARCHAR AS
  TO_CHAR(TRY_TO_DATE(VALUE:c6::VARCHAR,'YYYY-MM-DD'),'DD-MON-YYYY'),
  JOB_ID       VARCHAR AS (VALUE:c7::VARCHAR),
  SALARY       VARCHAR AS (VALUE:c8::VARCHAR),
  COMMISSION_PCT VARCHAR AS (VALUE:c9::VARCHAR),
  MANAGER_ID   VARCHAR AS (VALUE:c10::VARCHAR),
  DEPARTMENT_ID VARCHAR AS (VALUE:c11::VARCHAR)
)
LOCATION=@AWS_ET_CSV_STAGE
PATTERN='*.employees.*[.].csv'
FILE_FORMAT = CSV_ET_FILEFORMAT;
```

```
SELECT * FROM EXT_EMPLOYEES;
```

```
SELECT * FROM EXT_EMPLOYEES_FORMAT_DATE;
```

```
DESC EXTERNAL TABLE EXT_EMPLOYEES_FORMAT_DATE;
```

```
CREATE OR REPLACE EXTERNAL TABLE
EXT_EMPLOYEES_FORMAT_DATE_EMPLOYEE_ID_COMMISSION_PCT
(
  EMPLOYEE_ID  NUMBER AS TRY_TO_NUMBER(VALUE:c1::VARCHAR),
  FIRST_NAME   VARCHAR AS (VALUE:c2::VARCHAR),
  LAST_NAME    VARCHAR AS (VALUE:c3::VARCHAR),
  EMAIL        VARCHAR AS (VALUE:c4::VARCHAR),
  PHONE_NUMBER VARCHAR AS (VALUE:c5::VARCHAR),
  HIRE_DATE    VARCHAR AS TO_CHAR(TRY_TO_DATE(CAST(GET(VALUE, 'c6') AS
  VARCHAR(16777216)), 'YYYY-MM-DD'), 'DD-MON-YYYY'),
  JOB_ID       VARCHAR AS (VALUE:c7::VARCHAR),
```

```

    SALARY      VARCHAR AS (VALUE:c8::VARCHAR),
    COMMISSION_PCT NUMBER(38,2) AS TRY_CAST(CAST(GET(VALUE, 'c9') AS
VARCHAR(16777216)) AS NUMBER(38,2)),
    MANAGER_ID   VARCHAR AS (VALUE:c10::VARCHAR),
    DEPARTMENT_ID VARCHAR AS (VALUE:c11::VARCHAR)
)
LOCATION=@AWS_ET_CSV_STAGE
PATTERN='.*employees.*[.]csv'
FILE_FORMAT = CSV_ET_FILEFORMAT;

```

```

SELECT * FROM EXT_EMPLOYEES;

```

```

SELECT * FROM EXT_EMPLOYEES_FORMAT_DATE_EMPLOYEE_ID_COMMISSION_PCT;

```

```

CREATE OR REPLACE EXTERNAL TABLE EXT_EMPLOYEES_NO_COLS
WITH LOCATION=@AWS_ET_CSV_STAGE
FILE_FORMAT = (FORMAT_NAME = CSV_ET_FILEFORMAT);

```

```

SELECT * FROM EXT_EMPLOYEES_NO_COLS;

```

```

SELECT * FROM @AWS_ET_CSV_STAGE;

```

```

SELECT
    T.$1,
    T.$2,
    T.$3,
    T.$4,
    T.$5,
    T.$6 ,
    T.$7,
    T.$8,
    T.$9,
    T.$10,
    T.$11
FROM @AWS_ET_CSV_STAGE T;

```